

Evaluating Access to Healthcare

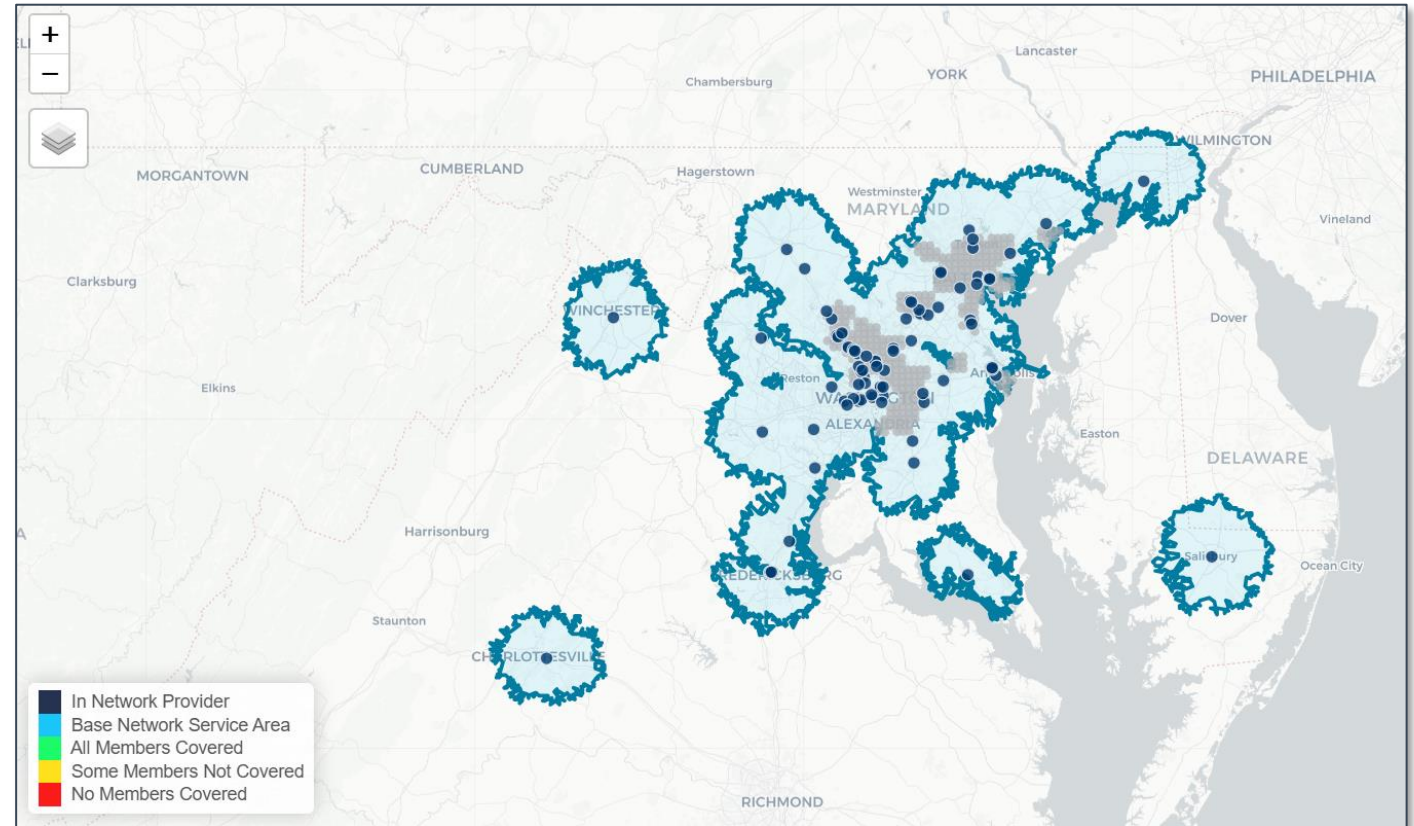
A modern spatial data workflow in R/python

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Healthcare Networks

- Network = Healthcare providers within a given state, county, insurance plan, etc
- Can be represented in space + time
- Physical “footprint” of accessible care



Why do we measure it?

- **Network Adequacy** = a health plan's ability to provide access to physicians/hospitals to meet the community's health care needs
- Evaluate physical barriers to healthcare access
- State/Federal regulations define standards of physician-to-enrollee ratio, time/distance, wait times

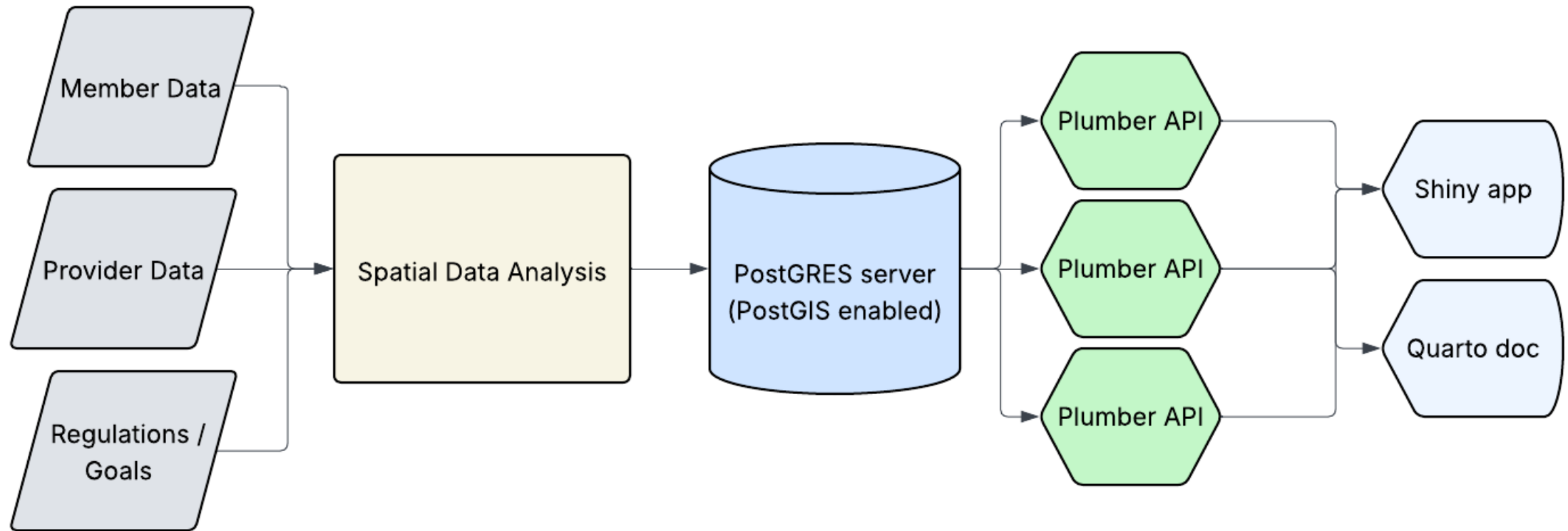
Table 1: Proposed Federal Marketplace Time/Distance Standards for Selected Specialties, 2023					
	Maximum Time and Distance Standards (minutes/miles) Per County Type				
Specialty Area	Large Metro	Metro	Micro	Rural	Counties with extreme access considerations
Primary Care	10/5	15/10	30/20	40/30	70/60
Cardiology	20/10	30/20	50/35	75/60	95/85
Emergency Medicine	20/10	45/30	80/60	75/60	110/100
Endocrinology	30/15	60/40	100/75	110/90	145/130



Solution Goals

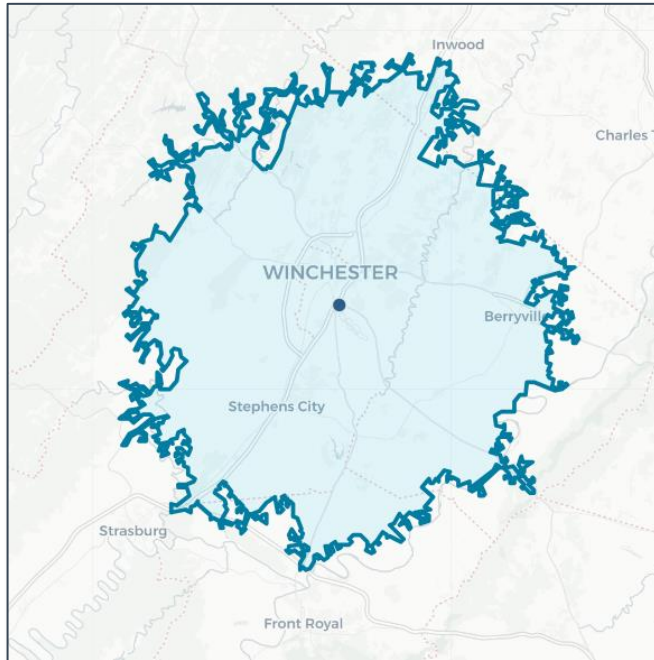
- Separation of concerns
(analysis, storage, deployment)
- Human scale geospatial analysis
- Exploration *and* Decision Support
- Reproducible
- Accessible

Workflow overview

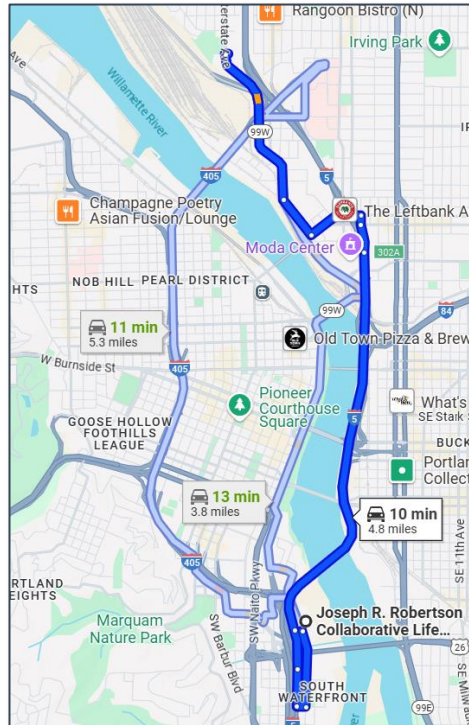


Data ingestion/Analysis

Service Area

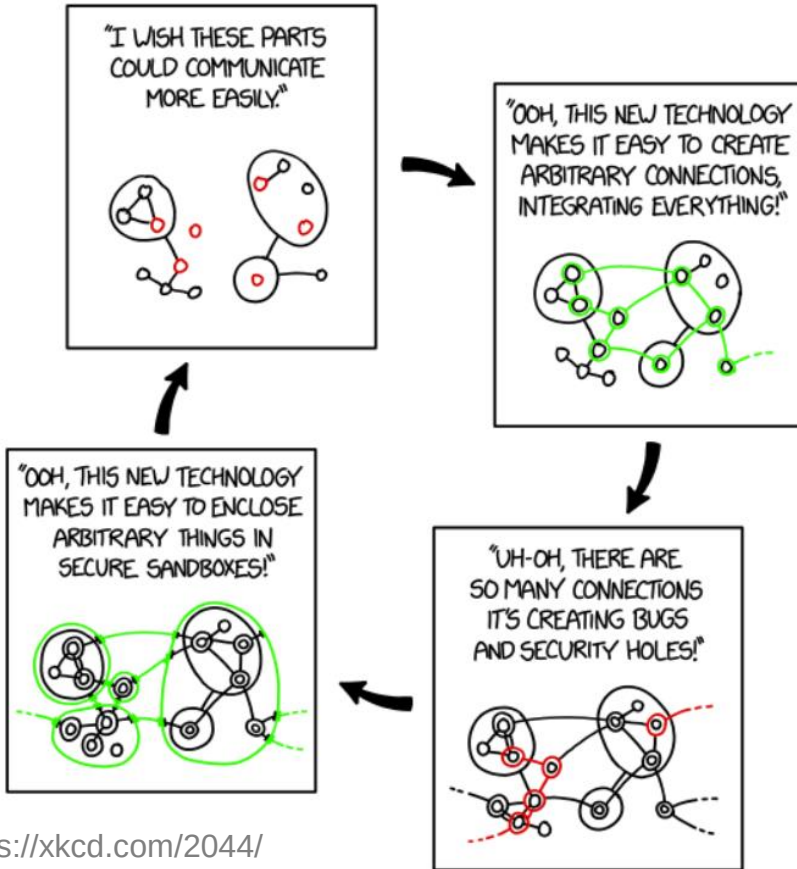


*Origin-Destination
analysis*



- Source data from internal data marts, APIs, flat files, screen grabs (oy)...
- Language agnostic
- ODCM vs Service Area
- ArcGIS Pro API optimized for python
- Internal R package wrapper (reticulate)
- StreetMapPremium

Storage



<https://xkcd.com/2044/>

On Prem
Resources

+





API endpoints

Utility Data

KP States

Oregon

Washington

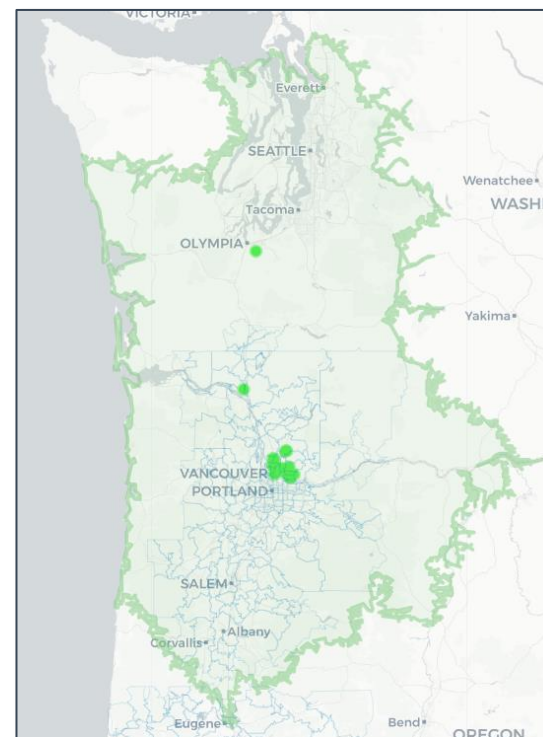
Colorado

...

Admin Layers



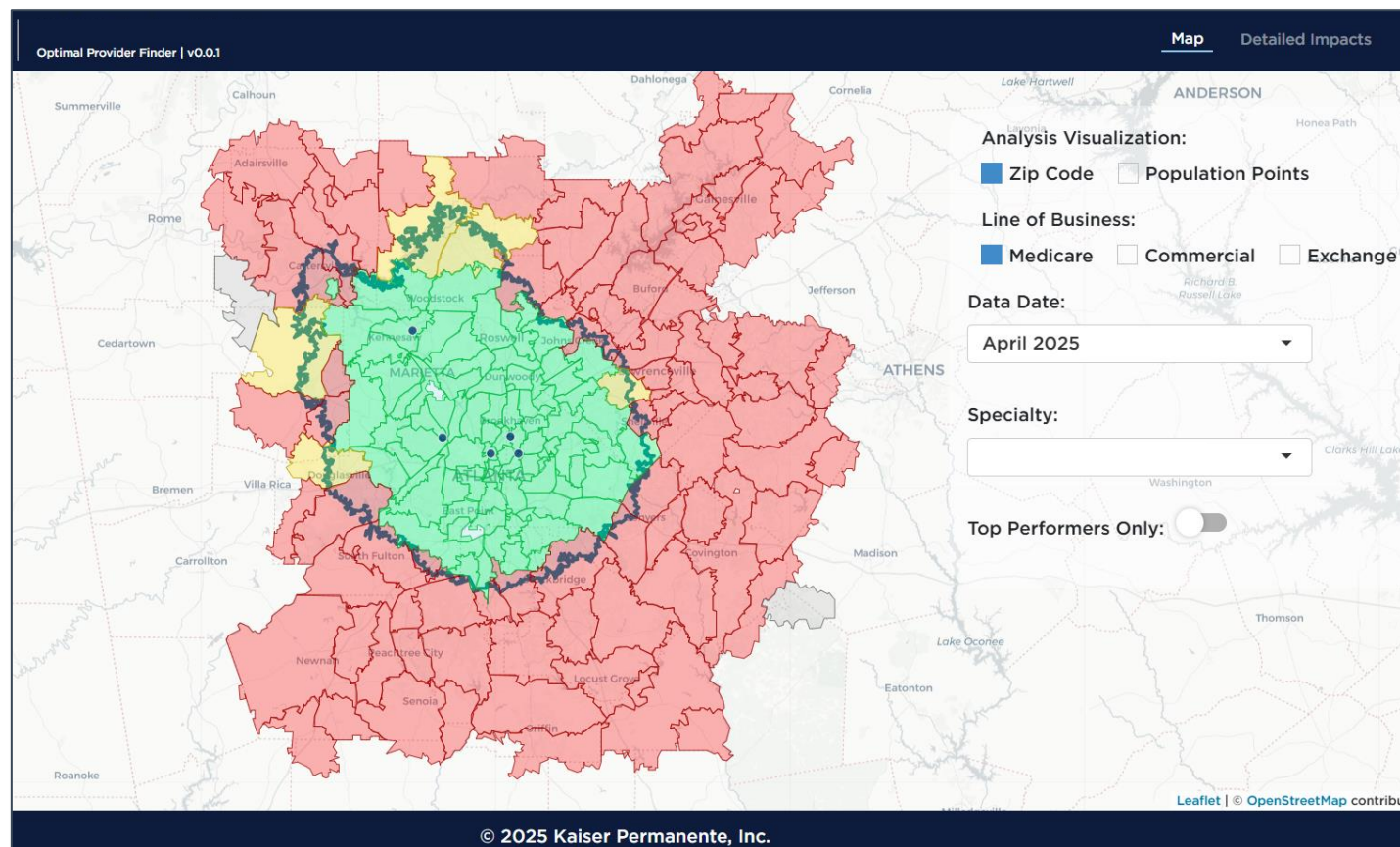
Analysis Results





Shiny Apps

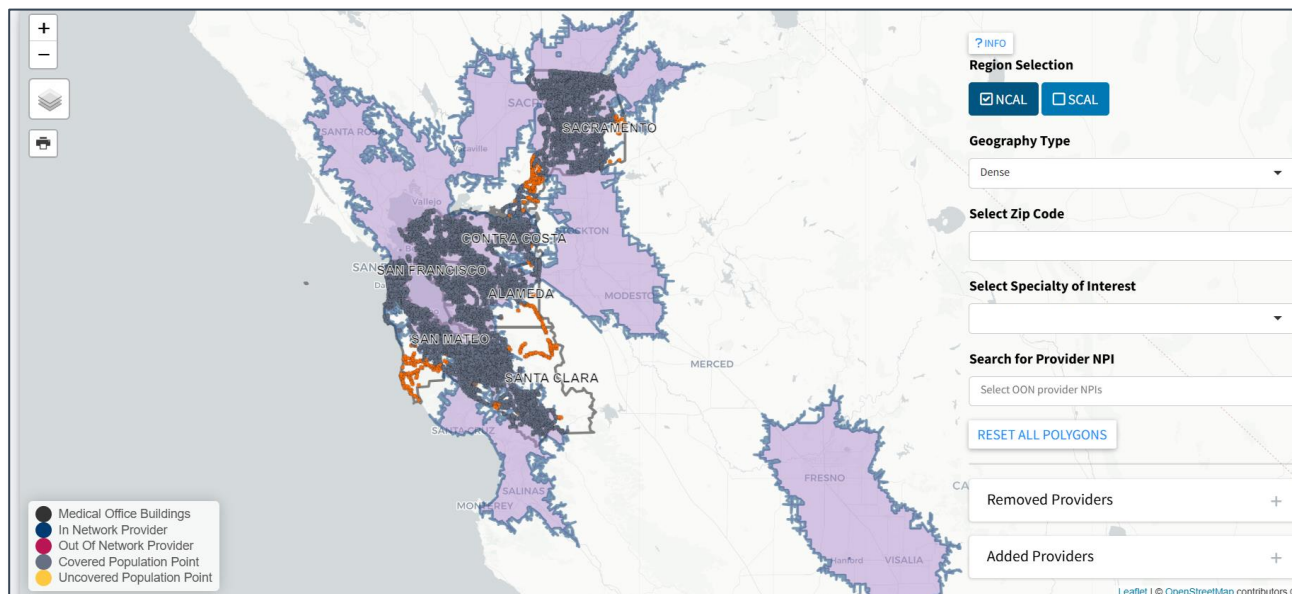
- Design freedom
- Open source extensibility
- Quick design+deploy+iterate timeline
- Integrates well with infrastructure
- Abstracted web design/dev complexity
- Easily shared





Shiny Apps

Geospatial exploration



Decision Support

Select Specialty

I

Filter Options

☐ Zips

☐ County

Select County

Alameda

Zip Code

95623

Select Provider for more information

	Zip	Projected Covered	Projected Uncovered	Baseline % Covered	Projected % Covered	% Change	Provider Name	Zips Affected
☐	95623	4530	142	96.96%	98.54%	1.58		2
☐	95623	4530	142	96.96%	99.02%	2.06		4
☐	95623	4530	142	96.96%	99.02%	2.06		4
☐	95623	4530	142	96.96%	99.02%	2.06		4
☐	95623	4530	142	96.96%	99.02%	2.06		4
☐	95623	4530	142	96.96%	99.02%	2.06		4



Markdown/Quarto Docs

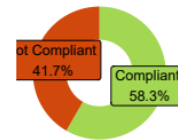
Network Adequacy for Specialists

NETWORK ADEQUACY RESULTS

Network Adequacy for Specialists

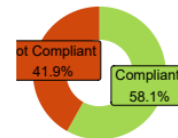
LOB	Product	Specialty	Compliance
Medicare	HMO	Endodontist	97.3%
		Multi Specialty	0%
		Oral Surgery	97.5%
		Periodontist	96.8%
		Prosthodontist	0%

Overall compliance for Medicare HMO



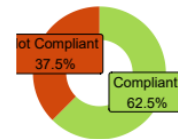
LOB	Product	Specialty	Compliance
Medicare	PPO	Endodontist	97%
		Multi Specialty	0%
		Oral Surgery	97.6%
		Periodontist	95.7%
		Prosthodontist	0.5%

Overall compliance for Medicare PPO



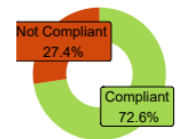
LOB	Product	Specialty	Compliance
QHP	HMO	Endodontist	100%
		Multi Specialty	0%
		Oral Surgery	100%
		Orthodontist	99.9%
		Pediatric Dentist	0%
		Pedodontist	100%
		Periodontist	100%
		Prosthodontist	0%

Overall compliance for QHP HMO



LOB	Product	Specialty	Compliance
QHP	PPO	Endodontist	100%
		Multi Specialty	0%
		Oral Surgery	100%
		Orthodontist	100%
		Pediatric Dentist	0%
		Pedodontist	100%
		Periodontist	100%
		Prosthodontist	80.9%

Overall compliance for QHP PPO



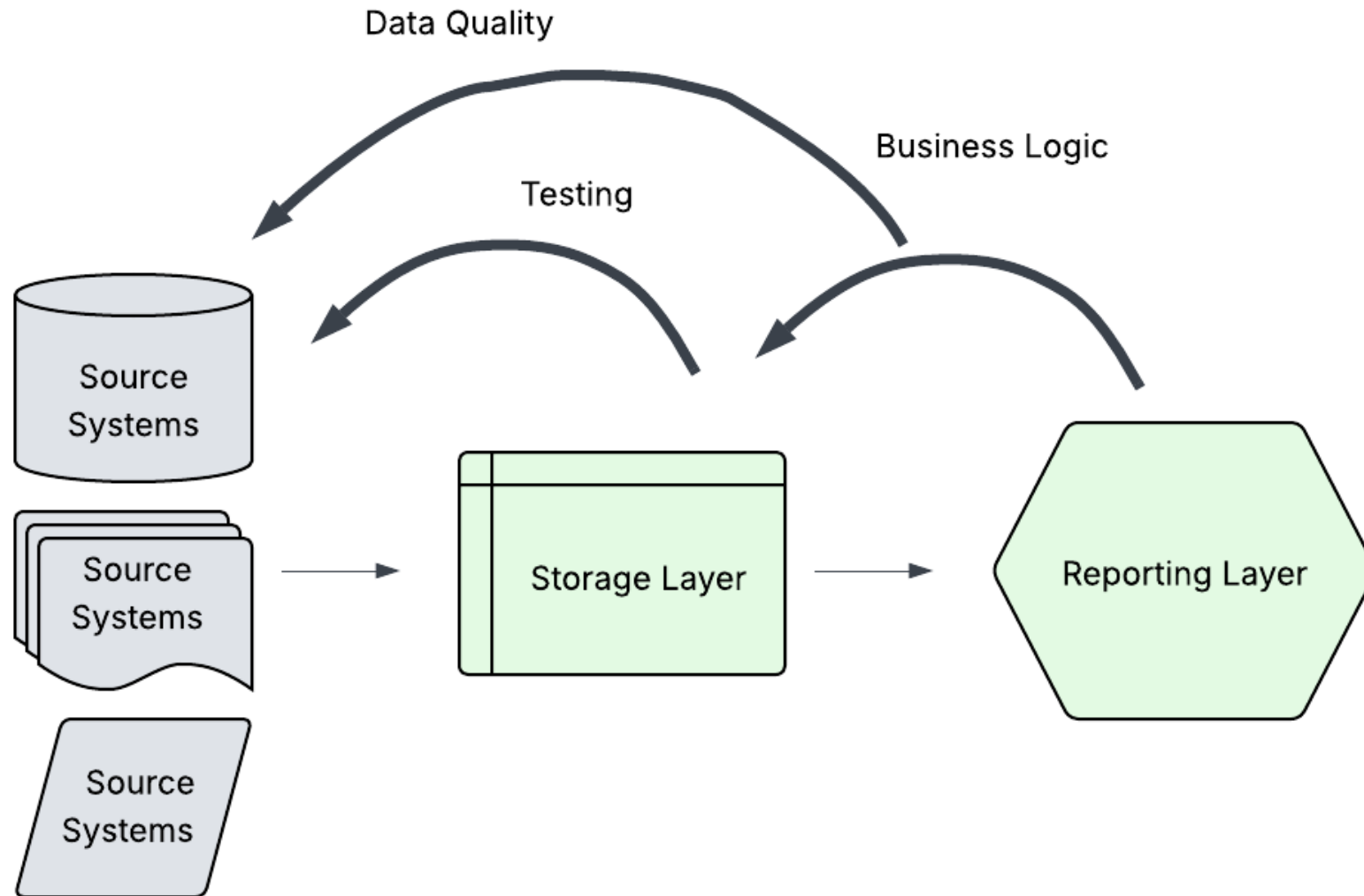
Network Adequacy Results by ZIP code

NETWORK ADEQUACY RESULTS

GA Delta Dental QHP PPO General Dentist (continued)

Vendor	LOB	Product	specialty	Market	ZIP	total_providers	Total Members	% Compliant
	QHP	PPO	General Dentist	GA	30633	1537	68	
	QHP	PPO	General Dentist	GA	30646	1537	32	
	QHP	PPO	General Dentist	GA	30621	1537	28	
	QHP	PPO	General Dentist	GA	30622	1537	48	
	QHP	PPO	General Dentist	GA	30677	1537	90	
	QHP	PPO	General Dentist	GA	30143	1537	133	
	QHP	PPO	General Dentist	GA	30148	1537	4	
	QHP	PPO	General Dentist	GA	30175	1537	36	
	QHP	PPO	General Dentist	GA	30177	1537	3	
	QHP	PPO	General Dentist	GA	30206	1537	21	
	QHP	PPO	General Dentist	GA	30256	1537	20	
	QHP	PPO	General Dentist	GA	30258	1537	21	
	QHP	PPO	General Dentist	GA	30292	1537	36	
	QHP	PPO	General Dentist	GA	30295	1537	29	
	QHP	PPO	General Dentist	GA	30217	1537	119	
	QHP	PPO	General Dentist	GA	30218	1537	23	
	QHP	PPO	General Dentist	GA	30222	1537	56	
	QHP	PPO	General Dentist	GA	30251	1537	14	
	QHP	PPO	General Dentist	GA	30293	1537	39	
	QHP	PPO	General Dentist	GA	31816	1537	44	
	QHP	PPO	General Dentist	GA	31830	1537	25	
	QHP	PPO	General Dentist	GA	30619	1537	14	
	QHP	PPO	General Dentist	GA	30630	1537	24	
	QHP	PPO	General Dentist	GA	30648	1537	31	
	QHP	PPO	General Dentist	GA	30667	1537	11	
	QHP	PPO	General Dentist	GA	30064	1537	242	
	QHP	PPO	General Dentist	GA	30082	1537	142	
	QHP	PPO	General Dentist	GA	30126	1537	210	
	QHP	PPO	General Dentist	GA	30127	1537	334	
	QHP	PPO	General Dentist	GA	30002	1537	33	
	QHP	PPO	General Dentist	GA	30032	1537	258	
	QHP	PPO	General Dentist	GA	30035	1537	109	
	QHP	PPO	General Dentist	GA	30083	1537	243	
	QHP	PPO	General Dentist	GA	30084	1537	248	
	QHP	PPO	General Dentist	GA	30087	1537	220	
	QHP	PPO	General Dentist	GA	30294	1537	272	
	QHP	PPO	General Dentist	GA	30317	1537	75	
	QHP	PPO	General Dentist	GA	30008	1537	121	

“Shift Left”





Thank you!

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